Algorithms and Sustainability

Lecture 3: Typical migration routes: clustering trajectories

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Tuesday, 27 October 2026

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Plan for today

1. Recap: comparing two routes
2. Which routes do young storks take? Clustering in movement data
3. k-center clustering of points: Gonzalez's algorithm
4. Clustering routes: (k, ℓ)-center with simplification
5. Other ways to cluster time series, and back to the storks

Recap: comparing two routes

Coupling of P and Q: both move forward, every position is matched.

measure	best coupling minimizes
Fréchet δ_{dF}	the **largest** distance
DTW	the **sum** of distances

Both by dynamic programming in $O(nm)$ time.

Today we also need the **triangle inequality**

$$d(P, R) \leq d(P, Q) + d(Q, R).$$

- Fréchet distance: **yes**
- DTW: **no**

Check with $a = (0), b = (1, 2), c = (2, 2, 2)$ and $d(x, y) = |x - y|$.

$\mathrm{DTW}(a, b) = 3$, $\mathrm{DTW}(b, c) = 1$, but $\mathrm{DTW}(a, c) = 6$.

Block 1 · Computational Movement Analysis

Week 3 · today: from similar routes to typical routes

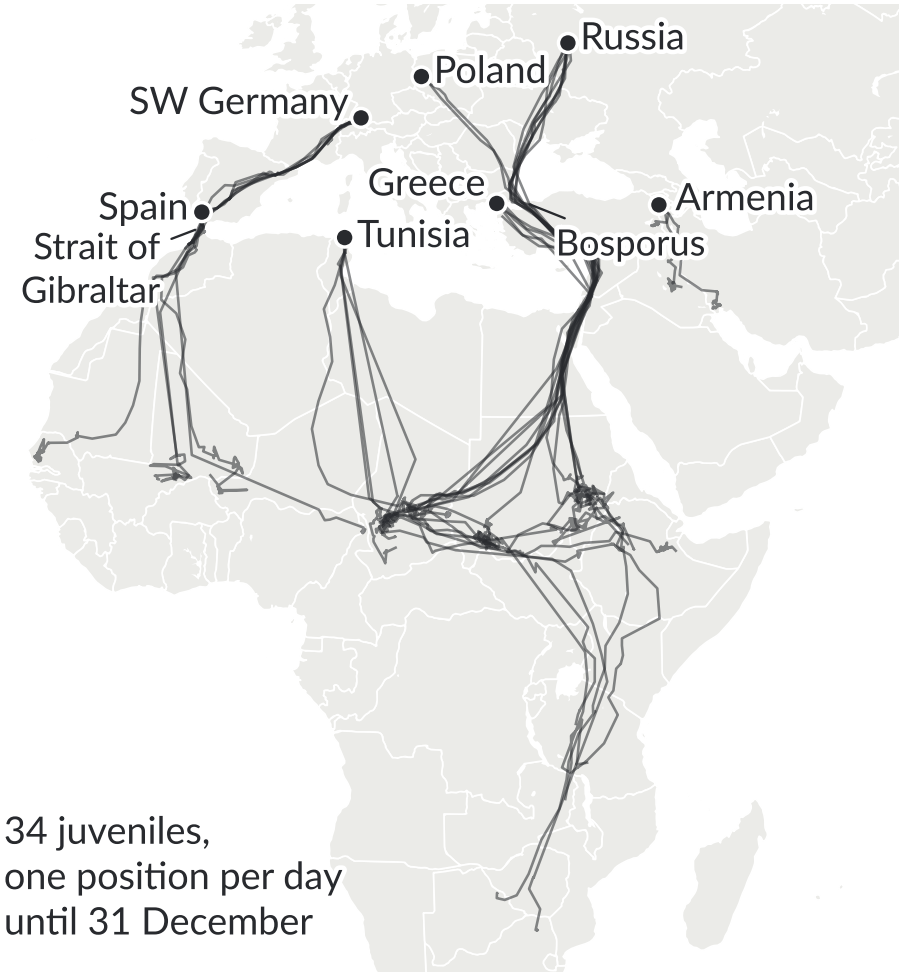
The question

1 · QUESTION

Which **routes** do young storks take, and **where** do they need protection?

- Migration concentrates at **bottlenecks**: Strait of Gibraltar, Bosporus
- Tracked large birds died more often from **human** than natural causes; **49%** of these deaths with known cause: power lines and wind turbines (Serratosa et al. 2024)
- Routes and wintering areas change (lecture 2)

34 first migrations



34 juveniles,
one position per day
until 31 December

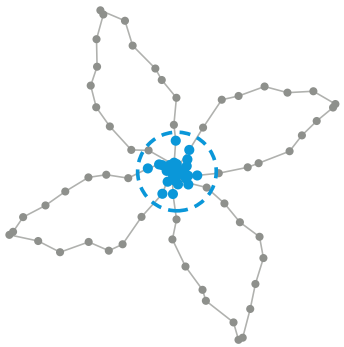
The juveniles of lecture 2 that **migrated** and were tracked **into the winter**: daily positions from departure to 31 December.

Which typical routes do you see? How many?

Goal: a few **representative routes** that summarize all 34.

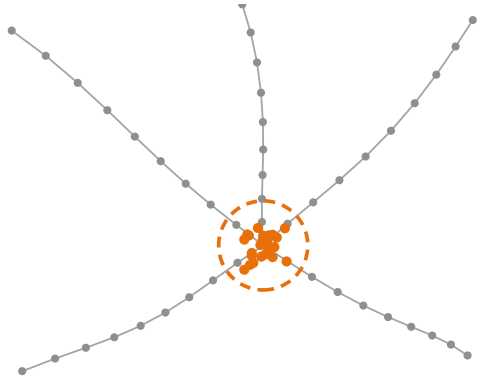
Data: Flack et al. (2016), Movebank, CC0. Map: Natural Earth.

Clustering in movement data



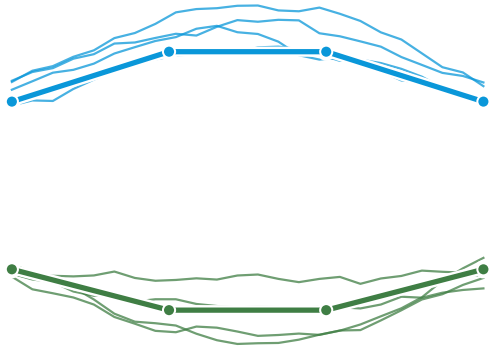
points of one track

nest, roost, dead bird



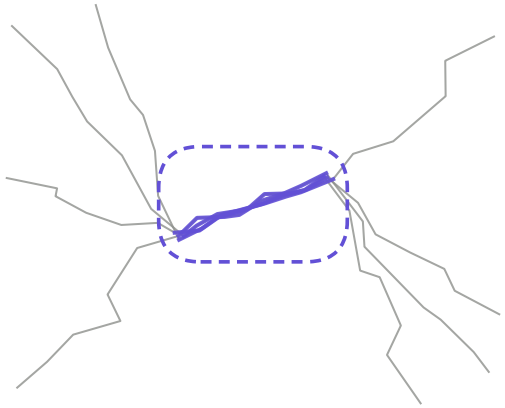
points of many tracks

vultures at a carcass



whole tracks

typical routes



parts of tracks

shared corridors

Points: where positions accumulate, in any order. **Curves:** the order matters.

Nests from revisits: Picardi et al. (2020). Carcasses from feeding vultures: Rast et al. (2024). Corridors: Buchin et al. (2011).

Quiz: points or curves?

For which question do we need to cluster curves rather than points?

A: Where did this stork build its nest?

B: Where do vultures feed on carcasses?

C: Which flyways do the juveniles use?

D: Where do the juveniles spend their first winter?

k-center clustering

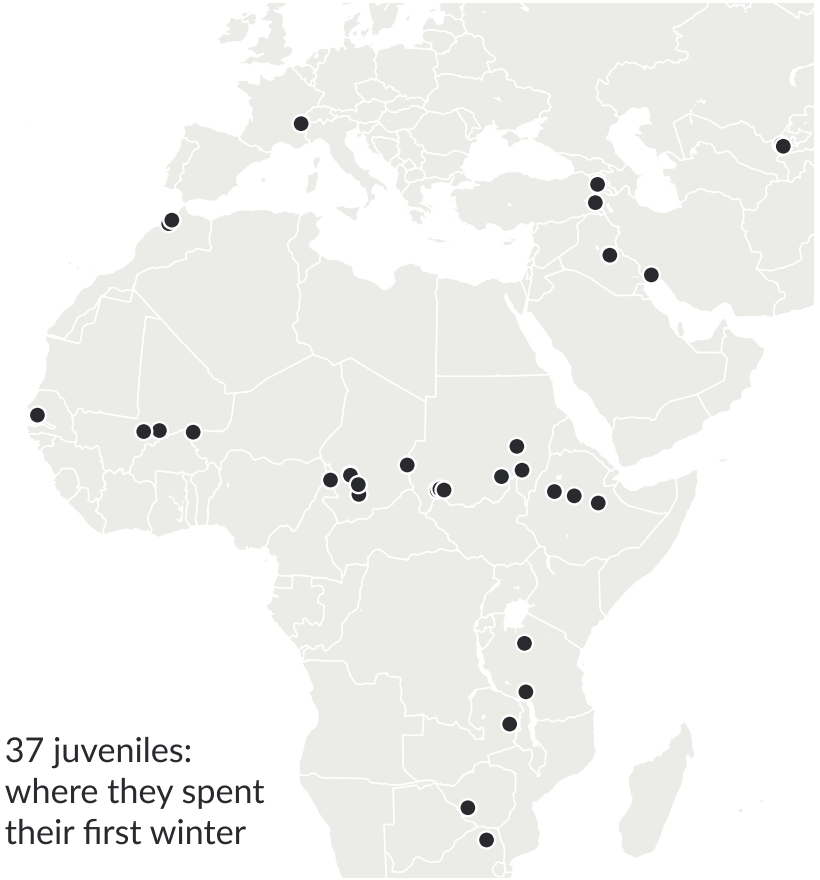
Definition

Objects P, distance d, number k: find **centres** C, $|C| = k$, minimizing the **radius**

$$r(C) = \max_{p \in P} \min_{c \in C} d(p, c).$$

No animal is badly represented. NP-hard.

Other objectives: k-median (sum), k-means (sum of squares).



37 juveniles:
where they spent
their first winter

Gonzalez's algorithm

Farthest first

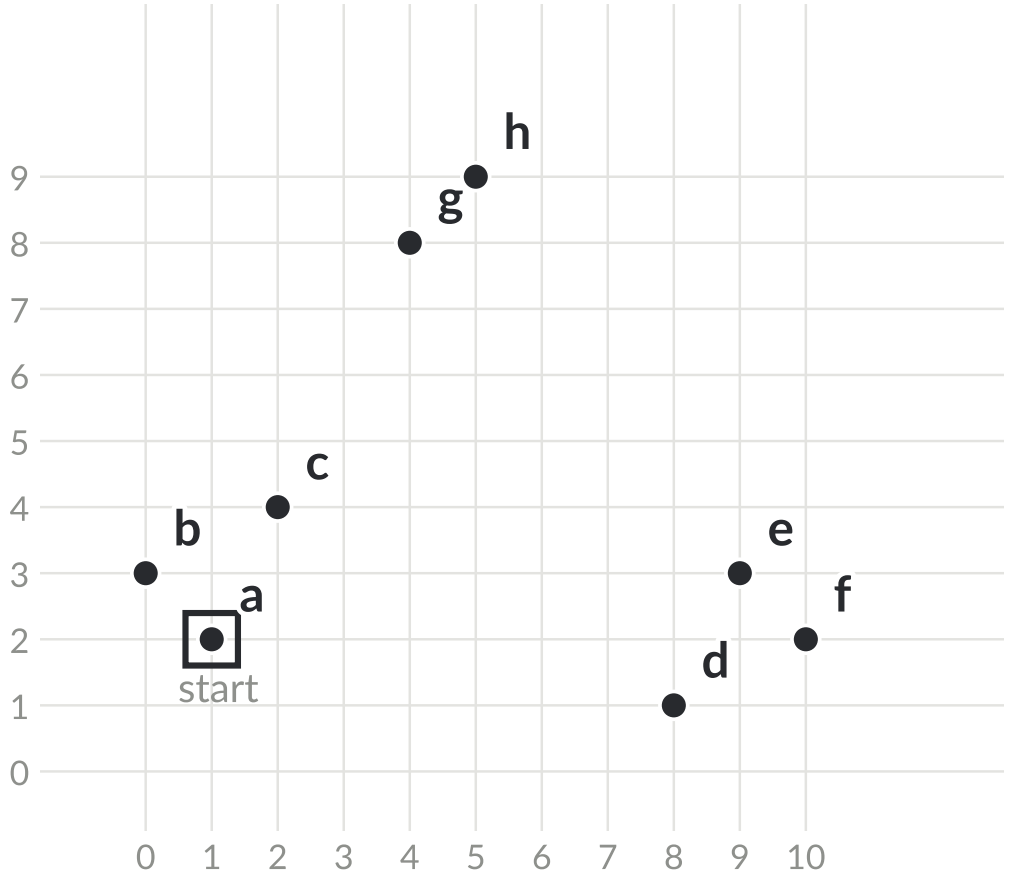
1. **Any** object: first centre
2. $k - 1$ times: the object **farthest** from its nearest centre becomes a centre
3. Assign objects to the nearest centre

$O(kn)$ time

Gonzalez (1985)



Your turn: Gonzalez by hand



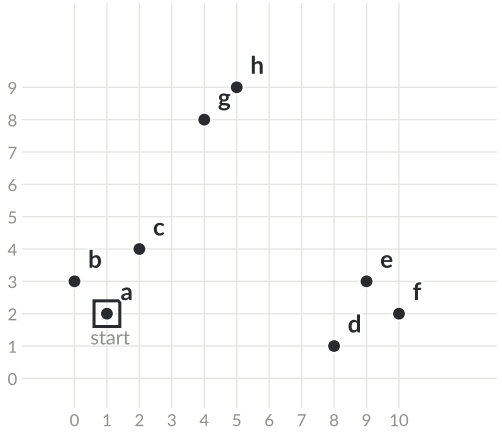
$k = 3$, start at a, Euclidean distance.

1. Which points become the **second** and **third** centre?
2. What is the **radius**?

Two minutes alone, one minute with your neighbour.

Done early? Is there a better choice of three centres?

Quiz: the third centre



Starting at a, which point does Gonzalez choose as the third centre?

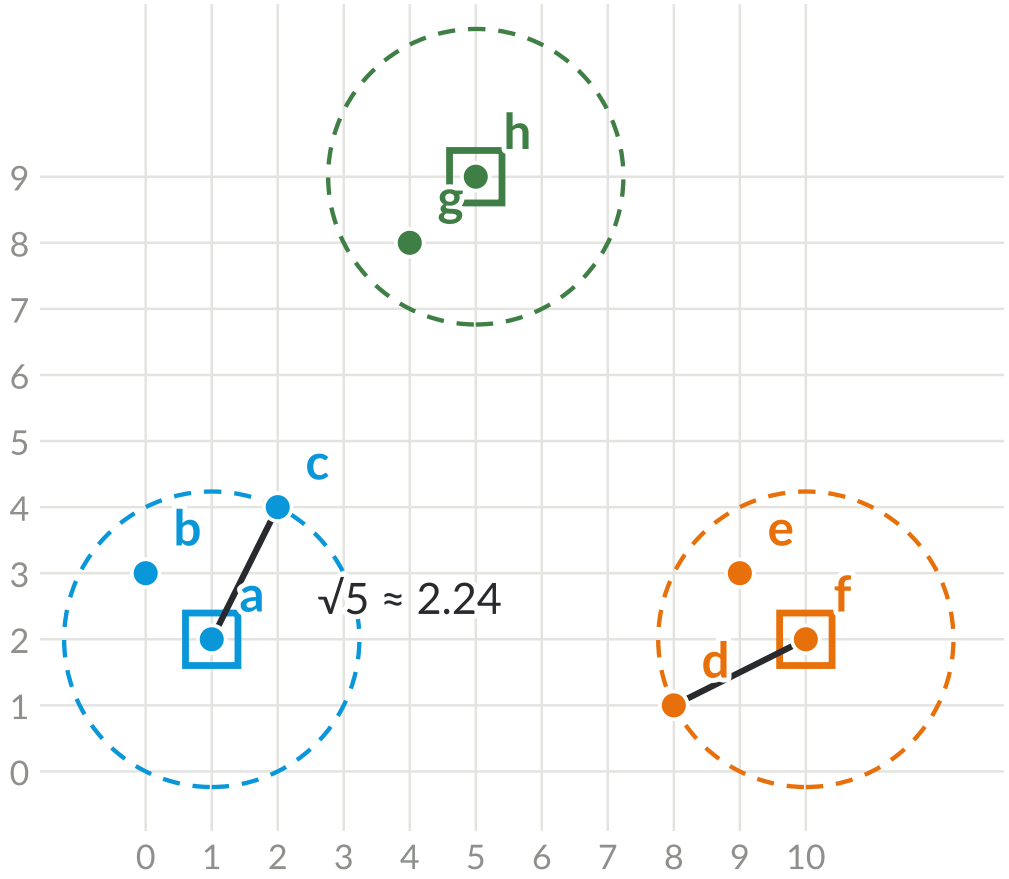
A: e

B: g

C: h

D: c

Solution



- Centres a, f, h
- Radius $\sqrt{5} \approx 2.24$ (c and d)
- Best centres anywhere: radius ≈ 1.18

Gonzalez is a factor **1.9** worse here. Can it be worse than a factor 2?

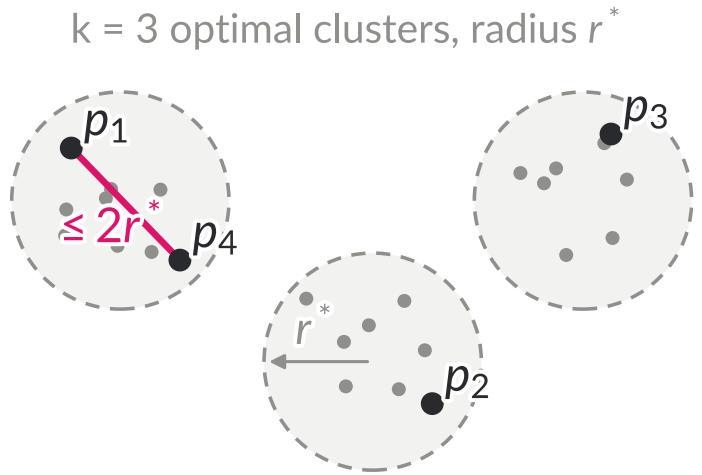
Why a 2-approximation?

Theorem (Gonzalez 1985)

Gonzalez's radius r is at most $2r^*$ (optimal radius r^*).

Proof. p_{k+1}: farthest object after k steps.

- $c_1, \dots, c_k, p_{k+1}$ are pairwise $\geq r$ apart
- Two lie in one optimal cluster
- Their distance is $\leq 2r^*$



Why pairwise $\geq r$? Where do we use the triangle inequality?

Can we do better?

- **Tight:** points $0, 1, 2$ on a line, $k = 1$: starting at 0 gives radius 2, the optimum is 1
- For general metric spaces, a $(2 - \varepsilon)$-approximation in polynomial time would imply P = NP (Hsu and Nemhauser 1979)
- **On the winter sites** ($k = 4$): Gonzalez 2,424 km; best four centres among the sites (all 66,045 choices): 2,302 km

> **In practice**
>
> Often close to the best possible, with a guarantee for every input. But the first centres are **extreme** objects.

From points to routes

Gonzalez needs only **distances** and the **triangle inequality**: use routes with the **discrete Fréchet distance**.

- Again a 2-approximation, but a centre is then **one bird's route**, with every detour

(k, ℓ)-center clustering

Find k curves C with **at most ℓ vertices** each, minimizing $\max_P \min_{c \in C} \delta_{\mathrm{dF}}(P, c)$.

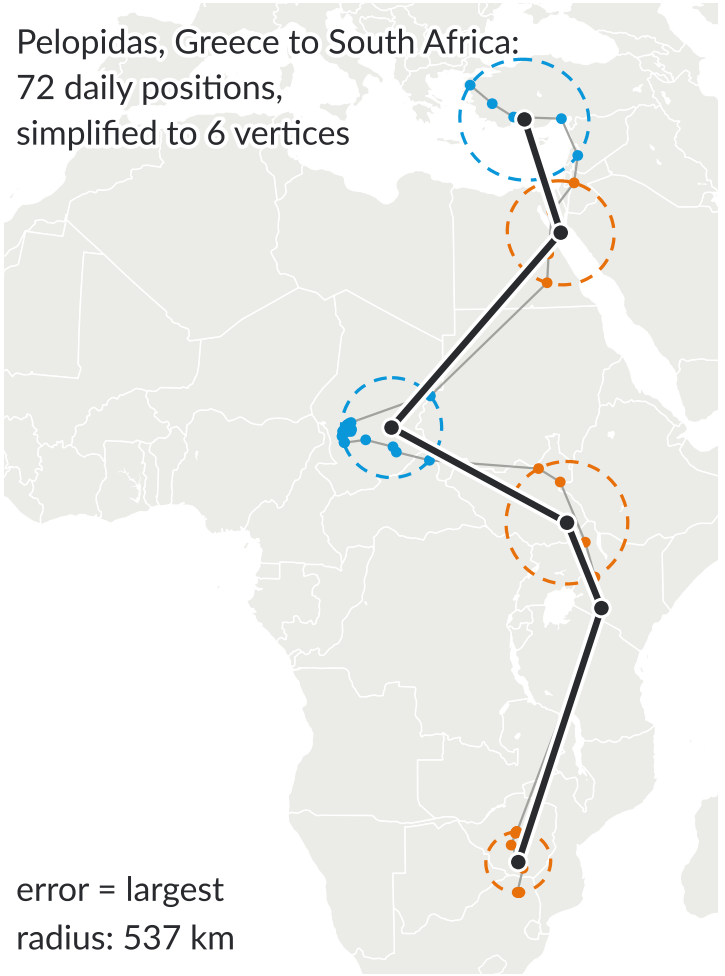
NP-hard; no approximation better than about 2.6 (Buchin et al. 2019).

Simplifying a route

Best curve with **at most ℓ vertices**, discrete Fréchet:

- Each vertex matches a **group of consecutive positions**
- Best vertex: centre of the group's **smallest enclosing circle**
- Error: the **largest radius**

Polynomial time (Bereg et al. 2008).



Gonzalez + simplification

1. Any route p_1; centre q_1 = its best simplification
2. Repeat: p_i = route **farthest** from its nearest centre; q_i = its simplification
3. Assign routes to the nearest centre

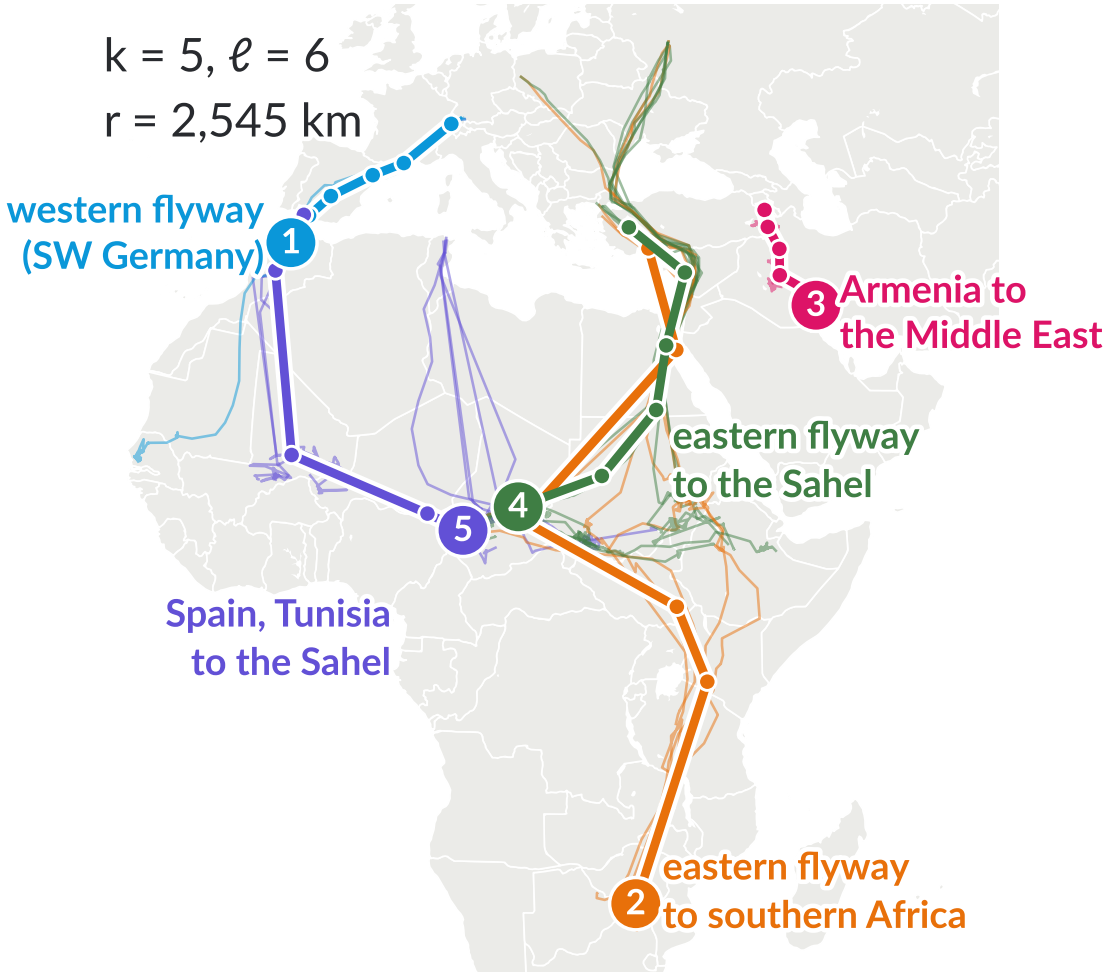
Theorem (Buchin et al. 2019)

Discrete Fréchet, any dimension: **3-approximation**, $O(kn \cdot \ell m \log(\ell + m))$ time (n routes, m positions each).

Why 3?

- $\delta(p_i, q_i) \leq r^*$: optimal centres have $\leq \ell$ vertices
- $\delta(p_i, p_j) \geq \delta(p_j, q_i) - \delta(q_i, p_i) \geq r - r^*$ for $i < j \leq k+1$
- Two share an optimal cluster: $r - r^* \leq 2r^*$

Typical routes of the young storks



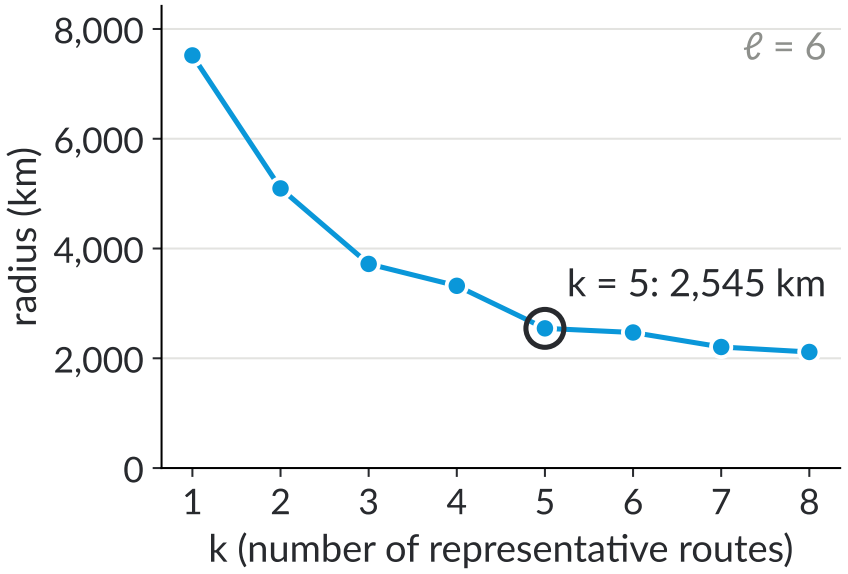
$k = 5$, $\ell = 6$, start at Karamell:

1. Karamell (SW Germany)
2. Pelopidas (Greece)
3. BraveHeart (Armenia)
4. Silvia (Greece)
5. C708 (Spain)

Radius: 7,518 → 5,095 → 3,719 → 3,321 → **2,545 km**

Data: Flack et al. (2016), Movebank, CC0.

How good, and which k?



- **Lower bound:** two of the 6 chosen routes share an optimal cluster, so r^* $\geq$ half their distance $\geq$ 1,133 km
- So 2,545 km is within a factor **2.25** of the optimum on these data
- $\ell = 4, 6, 8$: 2,621, 2,545, 2,470 km

Which k would you choose, and why?

Other ways to cluster time series

Approach	Representative	Example
centre-based, simple centres	curve with $\leq \ell$ vertices	(k, ℓ)-center or median, Fréchet or CDTW
partitional with medoids	an input series	k-medoids with DTW
partitional with averages	an average series	k-means with **DBA** (DTW barycenter averaging)
hierarchical	a tree of merges	single or complete linkage on pairwise distances
density-based	dense regions	DBSCAN (vultures), subtrajectory clusters
feature- or model-based	features, models	speed and distance with k-means; mixtures of HMMs

Reviews: Liao (2005), Aghabozorgi et al. (2015).

Back to the storks

4 · PRACTICE

What the routes show

- Western, central and eastern routes
- Where many birds pass: e.g. where to retrofit power lines

What they do not show

- Origin and destination mixed: for corridors, cluster **parts**
- **Extreme** routes first: outliers
- 34 juveniles, one year

Clustering summarizes data

Protection also needs mortality, habitat and threat data.

Next steps

- **Wednesday, 28 October**, 10:15, OH12 3.032: guest talk by **Andrea Kölzsch** (MoveApps) in the tutorial
- **From 3 November:** tutorials on Tuesdays, 14:15–15:45, OH14 E02
- **Tuesday, 3 November:** segmentation: when does an animal migrate, rest or forage?
- Project idea: a **MoveApp** that finds typical routes or shared corridors

Additional material

For self-study and the tutorial

DBA: averaging under DTW

DTW barycenter averaging (Petitjean et al. 2011):

1. Start with one series A as the average
2. Align every series to A with DTW
3. Replace each element of A by the mean of all elements aligned to it
4. Repeat until A no longer changes

- A heuristic that may stop in a local optimum
- An optimal average (1-median) is NP-hard, also under DTW (Buchin, Driemel and Struijs 2020)
- k-means with DBA: assign, then recompute the averages

k-center with outliers

One GPS error far away from all other positions becomes a centre in Gonzalez's algorithm.

k-center with z outliers

Find k centres that minimize the radius when the z farthest objects may be ignored.

- 3-approximation (Charikar et al. 2001)
- In practice: clean the data first (lecture 2 and next week: segmentation)

Why is (k, ℓ)-center hard?

- With $k = 1$ and ℓ unrestricted, any input curve is a 2-approximation: $\delta(P, Q) \leq \delta(P, c^*) + \delta(c^*, Q) \leq 2r^*$
- Computing the optimal centre is nevertheless NP-hard
- Idea of a reduction: from the **shortest common supersequence** of strings over $\{A, B\}$; encode each string as a curve so that a centre curve corresponds to a common supersequence

Buchin et al. (2019), Section 3

Shared corridors: subtrajectory clusters

A **subtrajectory cluster** $C(m, \ell, d)$: m disjoint parts of the trajectories, each of length ℓ, with pairwise Fréchet distance at most d.

- Deciding whether a cluster exists is NP-hard (reduction from maximum clique)
- A 2-approximation in d: use one part as reference curve; sweep over the free-space diagram

Buchin, Buchin, Gudmundsson, Löffler and Luo (2011)

Check your understanding

1. Find an input where Gonzalez's first centre changes the clusters.
2. Why does the 2-approximation proof fail for DTW?
3. Where does the 3-approximation proof use an **optimal** simplification?
4. Why allow $k\ell$ vertices in total instead of ℓ per centre?

Possible answers

1. Points $0, 4, 5, 9$, $k = 2$: start at 0 gives $\{0, 4\}, \{5, 9\}$, start at 4 gives $\{0, 4, 5\}, \{9\}$. 2. No triangle inequality. 3. $\delta(p_i, q_i) \leq r^*$. 4. Simple clusters need few vertices.

References: movement and conservation

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References: time series

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- Unless credited otherwise, figures: own work, CC BY 4.0, made with `tools/make_clustering_figures.py`. The stork figures are based on data from Flack et al. (2015), Movebank Data Repository, CC0; maps made with Natural Earth (public domain).
- The schematic of four kinds of clustering and the exercise use constructed data, not measurements.
- Made with Decker: https://github.com/decker-edu/decker